Forces and motion

45 questions: 15 multiple choice, 30 structured.

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1 · Part 1 Paper 1 & 2 Units [1]

Which unit is the unit of gravitational field strength?

A N/kg

B N

C kg/N

D m/s

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2 · Part 1 Paper 2 only Units [1]

Which quantity is measured in newton metres (Nm)?

A force

B momentum

C moment of a force

D pressure

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3 · Part 1 Paper 1 & 2 Movement and position [1]

In a race, a cyclist travels 4.5 km in 15 minutes.

What is the average speed of the cyclist?

A 0.30 m/s

B 18 m/s

C 300 m/s

D 5.0 m/s

4 · Part 1

Paper 1 & 2 | Movement and position | [1]

The velocity of a motorcycle increases uniformly from 6.0 m/s to 24 m/s. Its acceleration is 3.0 m/s^2.

What is the time taken for this change in velocity?

A 8.0 s

B 6.0 s

C 2.0 s

D 54 s

5 · Part 1

Paper 1 & 2 | Movement and position | [1]

Starting from rest, a dog runs in a straight line. Its velocity–time graph is a straight line from 0 m/s at time 0 s to 6.0 m/s at time 5.0 s.

What is the distance travelled by the dog in the 5.0 s?

A 15 m

B 30 m

C 1.2 m

D 0.83 m

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6 · Part 1

Paper 1 & 2 Units [6]

One student writes a report on an experiment with a trolley. The student forgets to include the units of the measurements.

(a) Complete the table by giving the unit of each quantity. Use unit symbols.

quantity	unit
mass of the trolley	
force on the trolley	
velocity of the trolley	
time taken	

[2]

(b) In another part of the report, the student gives a speed of 18 km/h.

Show that 18 km/h is equal to 5 m/s. [2]

(c) Deduce the unit of acceleration from the units of velocity and time. Explain your answer. [2]

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7 · Part 1

Paper 2 only

Units [4]

Four students each measure the momentum of a toy trolley moving along a track. Amir, Bella and Chen measure the same trolley. Dina measures a different trolley. They record their results in different units.

student	momentum recorded
Amir	0.36 kg m/s
Bella	360 g m/s
Chen	36 kg cm/s
Dina	4.5×10^4 g cm/s

(a) Show that Bella's value is equal to Amir's value. [1]

(b) Calculate Chen's value in kg m/s. [1]

(c) Deduce whether Dina's trolley has more momentum or less momentum than Amir's trolley. [2]

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8 · Part 1

Paper 1 & 2 Movement and position [9]

The table describes a bus journey between two towns. The journey has three stages.

stage	what happens
1	the bus travels 6.0 km in 10 minutes
2	the bus stops at a bus station for 5.0 minutes
3	the bus travels 9.0 km in 15 minutes

(a) State the equation linking average speed, distance moved and time taken. [1]

(b) Calculate the average speed of the bus during stage 1. Give your answer in m/s. [3]

(c) Calculate the average speed of the bus for the whole journey, including the stop. Give your answer in m/s. [3]

(d) Explain why the average speed for the whole journey is less than the average speed during stage 3. [2]

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9 · Part 1

Paper 1 & 2 Movement and position [9]

One student investigates how the height of a ramp affects the average speed of a toy car as it rolls down the ramp.

The student has a toy car, a wooden ramp 1.2 m long, a pile of books, a metre rule and a stopwatch. One end of the ramp rests on the floor and the other end rests on the pile of books.

(a) Identify the independent variable and the dependent variable in this investigation. [2]

(b) State one variable that the student should keep the same. [1]

(c) Describe a method the student could use to carry out this investigation. Your answer should include how the student obtains reliable results. [6]

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10 · Part 1

Paper 1 & 2 Movement and position [8]

One student investigates the motion of a tennis ball rolling along a flat corridor floor.

She marks two lines on the floor 5.0 m apart. She rolls the ball towards the lines in the same way each time. Her friend uses a stopwatch to time how long the ball takes to travel from the first line to the second line.

The table shows the results of five trials.

trial	time / s
1	2.4
2	2.6
3	3.9
4	2.5
5	2.5

(a) Identify the anomalous result. [1]

(b) Calculate the mean time, ignoring the anomalous result. [2]

(c) Calculate the average speed of the ball between the two lines. [2]

(d) Suggest one possible cause of the anomalous result. [1]

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(e) Explain why timing the ball over 5.0 m gives a more accurate value of its average speed than timing it over 0.50 m. [2]

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11 · Part 1

Paper 1 & 2 | Movement and position | [8]

In a race, a sprinter starts from rest. After 2.0 s her velocity is 8.4 m/s.

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(a) State the equation linking acceleration, change in velocity and time taken. [1]

(b) Calculate the average acceleration of the sprinter during the first 2.0 s. [2]

(c) A second sprinter also starts from rest. She reaches a velocity of 8.4 m/s after 2.4 s.

Explain which sprinter has the greater average acceleration. [2]

(d) The first sprinter crosses the finish line at a velocity of 10.5 m/s. After she crosses the line, her velocity decreases uniformly. Her acceleration is −1.5 m/s^2.

Calculate her velocity 3.0 s after she crosses the finish line. [3]

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12 · Part 1

Paper 1 & 2 Movement and position [8]

The table shows the velocity of a train at different times after it leaves a station. The train moves along a straight track.

time / s	velocity / m/s
0	0
10	6.0
20	12.0
30	18.0
40	18.0
50	18.0

The velocity–time graph of these results is a straight line from 0 s to 30 s, followed by a horizontal straight line from 30 s to 50 s.

(a) State what the gradient of a velocity–time graph represents. [1]

(b) Determine the acceleration of the train between 0 s and 30 s. [3]

(c) Determine the acceleration of the train between 30 s and 50 s. Give a reason for your answer. [2]

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(d) The velocity–time graph of a second train is also a straight line from 0 s to 30 s, but it has a steeper gradient.

Explain what this shows about the motion of the second train compared with the first train. [2]

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13 · Part 1

Paper 1 & 2 Movement and position [9]

The table gives the velocity of an electric scooter at four times as it moves along a straight path. The velocity–time graph of the journey is made of straight lines joining these points in order.

time / s	velocity / m/s
0	0
10	8.0
40	8.0
50	0

(a) State what the area between a velocity–time graph and the time axis represents. [1]

(b) Calculate the total distance travelled by the scooter in the 50 s. [4]

(c) Calculate the average speed of the scooter for the whole journey. [2]

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(d) Explain why the average speed for the whole journey is less than 8.0 m/s. [2]

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14 · Part 1

Paper 1 & 2 Movement and position [9]

On a straight, level road, a car travels at 15 m/s. The driver applies the brakes and the car slows down uniformly until it stops. The car travels 18 m while it is braking.

Use the equation

(final speed)^2 = (initial speed)^2 + (2 × acceleration × distance moved)

$v^2 = u^2 + (2 \times a \times s)$

(a) Calculate the acceleration of the car while it is braking. [3]

(b) Calculate the time taken for the car to stop. [2]

(c) The car now travels at 30 m/s. The driver brakes and the car has the same acceleration as in (a).

Show that the car travels 72 m while it is braking. [2]

(d) Explain why doubling the initial speed makes the distance travelled while braking four times larger. [2]

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15 · Part 1

Paper 1 & 2 | Movement and position | [9]

An aircraft starts from rest at one end of a runway. It accelerates uniformly in a straight line and reaches its take-off speed of 60 m/s after 24 s.

(a) Show that the acceleration of the aircraft is 2.5 m/s^2. [2]

(b) The equation linking final speed, initial speed, acceleration and distance moved is

(final speed)^2 = (initial speed)^2 + (2 × acceleration × distance moved)

$v^2 = u^2 + (2 \times a \times s)$

Use this equation to calculate the distance the aircraft travels along the runway before it reaches its take-off speed. [3]

(c) Calculate the average speed of the aircraft during these 24 s. [2]

(d) On another day the aircraft carries a heavier load. It needs the same take-off speed, but its acceleration is smaller.

Explain how this affects the distance the aircraft travels along the runway before take-off. [2]

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1 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Which list contains only vector quantities?

A distance, force, mass

B acceleration, force, velocity

C speed, time, weight

D displacement, energy, velocity

2 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

There is a weather satellite in orbit around the Earth.

Which type of force from the Earth keeps the satellite in its orbit?

A electrostatic

B frictional

C magnetic

D gravitational

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3 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

Three horizontal forces act on a box. The forces are 25 N to the left, 40 N to the right and 10 N to the left.

What is the resultant force on the box?

A 5 N to the right

B 5 N to the left

C 30 N to the right

D 75 N to the right

4 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

The trolley in this question has a mass of 2.5 kg. It is pulled along a level bench by a horizontal force of 12 N. A frictional force of 2.0 N acts on the trolley.

What is the acceleration of the trolley?

A 5.6 m/s^2

B 4.8 m/s^2

C 4.0 m/s^2

D 0.25 m/s^2

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5 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

An object has a weight of 30 N on Earth, where the gravitational field strength is 10 N/kg. The object is taken to a planet where the gravitational field strength is 4.0 N/kg.

What are the mass and the weight of the object on that planet?

	mass / kg	weight / N
A	3.0	12
B	3.0	30
C	1.2	12
D	30	120

6 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

The driver of a car travelling at a steady speed sees a hazard and brakes.

Which change increases the thinking distance but does not change the braking distance?

A The road surface is icy.

B The car is carrying a heavy load.

C The car's brakes are worn.

D The driver is very tired.

7 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [1]

One student stretches a rubber band and then removes the stretching force.

Which observation shows that the rubber band has behaved elastically?

A The extension of the band was proportional to the force.

B The band returns to its original shape and length.

C The band stays slightly longer than it was at the start.

D The band became thinner while it was stretched.

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8 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [1]

The cricket ball in this question has a mass of 150 g. It moves at a velocity of 20 m/s.

What is the momentum of the ball?

A 3000 kg m/s

B 7.5 kg m/s

C 3.0 kg m/s

D 0.13 kg m/s

9 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [1]

There is a book resting on a table. The Earth pulls the book downwards with a gravitational force.

Which force forms a Newton's third law pair with this force?

A the gravitational force of the book pulling the Earth upwards

B the force of the table pushing the book upwards

C the force of the book pushing the table downwards

D the gravitational force of the Earth pulling the table downwards

10 · Part 2

Paper 2 only | Forces, movement, shape and momentum | [1]

The light rod in this question is pivoted at its centre. A vertical downward force of 6.0 N acts on the rod 0.30 m to the left of the pivot.

Which vertical downward force, acting 0.20 m to the right of the pivot, will balance the rod?

A 1.8 N

B 4.0 N

C 6.0 N

D 9.0 N

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11 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [5]

During a match, a tennis ball travels towards a player. The player hits the ball with a racket and the ball moves back over the net.

(a) Describe the effects of the force from the racket on the ball. [3]

(b) While the ball moves through the air, a force acts on it that opposes its motion. State the name of this force. [1]

(c) State the type of force that pulls the ball towards the ground. [1]

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12 · Part 2

Paper 1 & 2 Forces, movement, shape and momentum [6]

(a) State how a vector quantity differs from a scalar quantity. [1]

(b) Two teams take part in a tug-of-war. Team 1 pulls the rope to the left with a total force of 1450 N. Team 2 pulls the rope to the right with a total force of 1320 N.

Calculate the size of the resultant force on the rope and state its direction. [2]

(c) Team 2 increases its pull until the horizontal forces on the rope are balanced. Team 1 still pulls with 1450 N.

State the total force that Team 2 now pulls with. [1]

(d) A runner completes exactly one lap of a 400 m running track. She finishes at the point where she started.

Explain why the distance she travels is 400 m but her displacement is zero. [2]

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13 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

The car in this question has a mass of 1200 kg. It starts from rest and accelerates uniformly along a straight, level road. After 6.0 s its velocity is 15 m/s.

(a) Calculate the acceleration of the car. [2]

(b) Calculate the resultant force on the car. [2]

(c) The total resistive force on the car is 600 N. Calculate the driving force on the car. [1]

(d) Explain why the driving force must increase if the car is to keep the same acceleration as its speed increases. [2]

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14 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

The small spacecraft in this question lands on the Moon. The spacecraft has a mass of 2000 kg. The gravitational field strength on the Moon is 1.6 N/kg.

(a) Explain why the mass of the spacecraft is the same on the Moon as on the Earth, but its weight is different. [2]

(b) Calculate the weight of the spacecraft on the Moon. [2]

(c) Just before landing, the spacecraft is moving downwards. Its rocket engine produces an upward force of 4000 N.

Calculate the acceleration of the spacecraft and state its direction. [3]

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15 · Part 2

Paper 1 & 2 Forces, movement, shape and momentum [8]

The table shows the thinking distance and the braking distance for a car on a dry road at three different speeds. The same driver is driving in each case.

speed / m/s	thinking distance / m	braking distance / m
10	6	8
20	12	32
30	18	72

(a) Calculate the stopping distance of the car at 20 m/s. [1]

(b) Describe how the thinking distance and the braking distance change as the speed increases. Use data from the table in your answer. [3]

(c) The car is now loaded with heavy luggage. The braking force is the same as before.

Explain why the braking distance at each speed is now greater. [2]

(d) The driver's reaction time is the same at all three speeds. Calculate the driver's reaction time. [2]

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16 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

The skydiver in this question has a mass of 75 kg. She jumps from a hovering helicopter. She falls for some time before opening her parachute. The gravitational field strength is 10 N/kg.

(a) Calculate the weight of the skydiver. [2]

(b) Explain why the skydiver accelerates at first but then falls at a terminal velocity. [4]

(c) State the size of the air resistance on the skydiver while she is falling at terminal velocity. [1]

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17 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

One student investigates the motion of a small steel ball falling through oil in a tall glass tube. Marks are placed on the tube every 10 cm below the surface of the oil. The student releases the ball at the surface and records the time when the ball passes each mark.

distance fallen / cm	time / s
0	0
10	0.40
20	0.70
30	0.95
40	1.20
50	1.45

(a) Explain how the results show that the ball reaches terminal velocity. [2]

(b) Calculate the terminal velocity of the ball in m/s. [2]

(c) The student repeats the experiment with the same ball in a much thicker oil.

Suggest how the terminal velocity of the ball changes. Give a reason for your answer. [2]

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(d) Suggest one way the student could measure the times more accurately. [1]

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18 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

One student investigates how the extension of a helical spring varies with the force applied to it. She has a spring, a clamp stand with a clamp, a metre rule and a set of slotted masses. Each slotted mass has a weight of 0.50 N.

(a) Describe a method the student could use to carry out this investigation. [4]

(b) State the independent variable and the dependent variable in this investigation. [2]

(c) State one safety precaution the student should take. [1]

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19 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [7]

One student hangs a spring from a clamp and measures its length as she increases the force on it.

force / N	length / cm
0	12.0
1.0	14.0
2.0	16.0
3.0	18.0
4.0	20.0
5.0	23.0
6.0	27.0

(a) Calculate the extension of the spring when the force is 3.0 N. [1]

(b) Identify the largest force in the table for which the spring obeys Hooke's law. Explain your answer using data from the table. [3]

(c) Predict the length of the spring when the force on it is 2.5 N. [1]

(d) The student removes the 6.0 N force. The length of the spring is now 13.5 cm.

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Explain what this shows about the behaviour of the spring. [2]

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20 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [6]

One student hangs weights from a rubber band and measures the extension of the band for each force.

force / N	extension / mm
0	0
2.0	3.0
4.0	8.0
6.0	15.0
8.0	25.0

(a) Explain whether the rubber band obeys Hooke's law. Use data from the table in your answer. [2]

(b) The student then wants to find out whether the width of a rubber band affects its extension for a given force.

State two variables she should keep the same. [2]

(c) Suggest one way the student could make her measurements of extension more accurate. [1]

(d) The student also tests a metal wire. For the same forces, the extensions of the wire are much smaller than those of the rubber band.

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Suggest why a long wire is used in this experiment. [1]

21 · Part 2

Paper 1 & 2 Forces, movement, shape and momentum [6]

The worker in this question pushes a crate of mass 50 kg across a level floor. She pushes horizontally with a force of 180 N. The frictional force on the crate is 130 N.

(a) State the direction of the frictional force on the crate. [1]

(b) Calculate the acceleration of the crate. [3]

(c) While the crate is moving, the worker reduces her push to 130 N. The frictional force stays at 130 N.

Describe the motion of the crate now. Give a reason for your answer. [2]

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22 · Part 2

Paper 1 & 2 Forces, movement, shape and momentum [6]

(a) Complete the table by stating the type of force acting in each situation.

situation	type of force
after being rubbed with a cloth, a plastic comb attracts small pieces of paper	
an apple falls from a tree to the ground	
a steel paper clip is pulled towards a magnet	
a book sliding across a table slows down and stops	

[4]

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(b) A car travels around a circular track at a constant speed.

Explain why there must be a resultant force on the car. [2]

23 · Part 2

Paper 1 & 2 | Forces, movement, shape and momentum | [6]

One student hangs different masses from a newtonmeter and records the readings.

mass / kg	weight / N
0.10	0.98
0.20	1.96
0.30	2.94
0.40	3.92
0.50	4.90

(a) State the equation linking weight, mass and gravitational field strength. [1]

(b) Use the results to determine the gravitational field strength. Give the unit. [2]

(c) The newtonmeter and the 0.40 kg mass are taken to a planet where the gravitational field strength is 25 N/kg.

Calculate the reading on the newtonmeter. [2]

(d) Suggest why the student used several different masses rather than just one. [1]

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24 · Part 2

Paper 2 only

Forces, movement, shape and momentum

[7]

The golf ball in this question has a mass of 0.045 kg and is at rest on the ground. A golf club hits the ball, and the ball leaves the club at a speed of 60 m/s. The club is in contact with the ball for 0.0015 s.

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(a) State the equation linking momentum, mass and velocity. [1]

(b) Calculate the momentum of the ball as it leaves the club. [2]

(c) Calculate the average force of the club on the ball.

Use the equation

force = change in momentum / time taken [2]

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(d) Golfers keep the club moving forward after hitting the ball. This is called "following through". It increases the time that the club is in contact with the ball.

Explain why following through makes the ball leave the club at a higher speed, for the same average force. [2]

25 · Part 2

Paper 2 only

Forces, movement, shape and momentum [8]

In a crash test, a car travelling at 15 m/s hits a wall and stops. A test dummy of mass 60 kg inside the car is brought to rest by its seat belt and an airbag.

(a) Calculate the change in momentum of the dummy. [2]

(b) The dummy takes 0.12 s to come to rest. Calculate the average force on the dummy.

Use the equation

$F = (mv - mu) / t$ [2]

(c) Explain how seat belts and airbags reduce the risk of injury to people in a car crash. [4]

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26 · Part 2

Paper 2 only

Forces, movement, shape and momentum [7]

Two ice skaters stand still on smooth ice, facing each other. Skater P has a mass of 60 kg and skater Q has a mass of 40 kg. They push against each other's hands and move apart. Immediately afterwards, skater Q moves to the right at 3.0 m/s. Friction from the ice is negligible.

(a) State the total momentum of the two skaters before they push. Give a reason for your answer. [2]

(b) Calculate the velocity of skater P immediately after they push apart and state its direction. [3]

(c) Use Newton's third law to explain why both skaters move, not just one of them. [2]

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27 · Part 2

Paper 2 only

Forces, movement, shape and momentum [8]

Railway truck X has a mass of 12 000 kg and moves at 2.5 m/s along a straight, level track. It collides with railway truck Y, which has a mass of 8000 kg. The two trucks join together and move off as one. Friction is negligible.

(a) In the first test, truck Y is stationary before the collision.

Calculate the speed of the joined trucks immediately after the collision. [3]

(b) In a second test, truck X again moves at 2.5 m/s. This time truck Y moves towards truck X at 1.0 m/s before the collision.

Calculate the velocity of the joined trucks immediately after the collision and state its direction. [3]

(c) During the collision, truck X loses momentum and truck Y gains the same amount of momentum.

Explain, in terms of the forces between the trucks, why this happens. [2]

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28 · Part 2

Paper 2 only

Forces, movement, shape and momentum [7]

The seesaw in this question is a uniform plank pivoted at its centre. One child, of weight 300 N, sits 1.6 m to the left of the pivot. A second child, of weight 400 N, sits on the right of the pivot so that the seesaw balances.

(a) State the principle of moments. [2]

(b) Calculate the distance of the second child from the pivot. [3]

(c) Explain why the weight of the plank does not affect whether the seesaw balances. [2]

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29 · Part 2

Paper 2 only

Forces, movement, shape and momentum [6]

The light plank in this question is 4.0 m long. It rests horizontally on two supports, one at each end. Support X is at the left end and support Y is at the right end. A painter of weight 600 N stands on the plank 1.0 m from support X.

(a) Calculate the upward force from support Y on the plank. [3]

(b) Calculate the upward force from support X on the plank. [1]

(c) The painter walks slowly along the plank from support X towards support Y.

Describe how the upward forces from the two supports change. [2]

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30 · Part 2

Paper 2 only

Forces, movement, shape and momentum

[6]

The gardener in this question uses a wheelbarrow. The axle of the wheel acts as the pivot. The total weight of the wheelbarrow and its load is 800 N. This weight acts through a point 0.45 m horizontally from the axle. The gardener lifts the handles with a vertical force that acts 1.2 m horizontally from the axle.

(a) State what is meant by the centre of gravity of an object. [1]

(b) Calculate the vertical force the gardener must apply to hold the wheelbarrow balanced with its legs just off the ground. [3]

(c) Explain, in terms of moments, why the force needed is smaller when the load is placed closer to the wheel. [2]

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